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COMPUTATIONAL MEDICINE ASSIGNMENT

BRAIN TUMOUR (HIGH GRADE GLIOMA) SURGICAL PLANNING

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Preamble

This report provides a clinical and computational approach to high-grade glioma diagnosis, MRI image analysis, tumour segmentation and surgical planning.

High-grade gliomas are primary brain tumours that present considerable challenges in diagnosis and surgical management. Their growth/progression patterns are often irregular, and tumour cells may infiltrate surrounding brain tissue beyond the boundaries that can be identified through conventional magnetic resonance imaging (MRI), making surgical planning particularly difficult. A neurosurgical team must therefore establish the extent of the tumour, identify its relationship with important neurological structures and determine how much tissue can be safely removed without causing significant neurological impairment.

Glioblastoma, IDH-wildtype, is one of the most aggressive adult-type diffuse gliomas. Its diagnosis and management require more than identifying the visible tumour on an MRI scan. Histopathological examination, molecular characterisation, assessment of the patient's clinical condition and detailed imaging analysis all, contribute to treatment decisions.

MRI is especially useful because it provides information about tumour anatomy, tissue characteristics, vascularity and surrounding brain structures without requiring an invasive procedure. When combined with image-processing and machine-learning techniques, MRI can also be used to delineate tumour regions, calculate tumour volumes and extract quantitative features that may support surgical planning.

This report examines three related aspects for the brain tumour surgical planning including:

1. The *appropriate biological sample and diagnostic investigations* for a patient with high-grade glioma.
2. MRI *imaging characteristics that may predict treatment outcomes* and influence surgical decisions.
3. The application of *medical image processing and machine learning* to brain tumour segmentation and quantitative feature extraction.

The practical component (image segmentation and feature extraction) uses multimodal MRI data from the Brain Tumour Segmentation (BraTS) research dataset to demonstrate a reproducible approach to tumour segmentation, visualisation and feature extraction.

APPROPRIATE BIOLOGICAL SAMPLE TYPE USED TO DIAGNOSE A PATIENT WITH HIGH-GRADE GLIOMA

The preferred diagnostic sample for a patient suspected of high-grade glioma, is the brain tissue, which is obtained through a stereotactic needle or surgical resection.

MRI imaging can only suggest the presence of high-grade glioma, but is insufficient to establish in definitive terms, its histological and molecular diagnosis. Tissue/biopsy examination is therefore required to identify the tumour type, determine its grade and establish its molecular characteristics.

Methods of Obtaining Biopsies:

1. **Stereotactic needle biopsy**

This involves collecting small tissue samples from the suspected tumour using a needle guided by preoperative MRI or CT images. A three-dimensional coordinate system helps the neurosurgeon direct the needle towards the selected region.

The method is used when tumour is located in a deep brain structure like the thalamus or brain-stem or when an extensive surgical removal would carry an unacceptable risk.

The main advantage is that it provides tissue for diagnosis without requiring the surgeon to remove a large portion of the tumour. However, a biopsy may provide limited information about the tumour's overall composition because high-grade gliomas are heterogeneous.

For example, a glioblastoma may contain regions of necrosis, densely packed tumour cells, abnormal blood vessels and infiltrating tumour tissue. A small biopsy could sample only one of these regions and fail to represent the complete tumour.

2. **Surgical Resection Specimen:**

This involves removing as much tumour tissue as can be safely achieved. The removed tissue is submitted for histopathological and molecular examination.

Compared with a needle biopsy, a resection specimen provides a larger quantity of tissue and may allow the pathologist to examine different tumour regions.

Potential clinical benefits:

- It provides material for a more comprehensive diagnosis.
- It reduces the amount of visible tumour, potentially improving subsequent disease management.

However, complete removal is not always possible. A tumour that extends into the motor cortex, language areas or critical white matter pathways may need to be partially resected to preserve neurological function.

The choice between biopsy and resection therefore depends on tumour location, patient condition, surgical accessibility and the anticipated risks of the procedure.

Histopathological Examination:

Once the sample has been obtained, it undergoes histopathological examination.

The tissue is generally fixed, processed, embedded in paraffin, sectioned and stained using haematoxylin and eosin (H&E). The stained sections are examined under a microscope.

Features for pathological identification in a high-grade glioma include:

- Increased cellularity.
- Nuclear atypia and cellular pleomorphism.
- Increased mitotic activity.
- Microvascular proliferation.

- Areas of tumour necrosis.

Microvascular proliferation and necrosis are particularly relevant morphological characteristics associated with glioblastoma.

Histopathology provides information about the microscopic appearance of the tumour. Nevertheless, contemporary diagnosis requires molecular testing alongside conventional morphology.

Making Molecular Investigations:

The 2021 World Health Organization (WHO) classification of the central nervous system tumours places considerable emphasis on molecular characteristics. Therefore, the diagnosis of adult diffuse gliomas is now based on an integrated assessment of histology and molecular findings.

Molecular marker or test	**Diagnostic or clinical significance**
IDH1 and IDH2	Helps distinguish IDH-mutant gliomas from IDH-wildtype gliomas.
MGMT promoter methylation	Provides information about the likelihood of benefit from temozolomide chemotherapy.
ATRX	Loss of ATRX expression supports certain molecular tumour classifications.
TP53	Mutations may be associated with particular diffuse glioma subtypes.
1p/19q codeletion	Helps identify oligodendroglioma when considered alongside IDH mutation.
EGFR amplification	Can support the diagnosis of glioblastoma, IDH-wildtype in the appropriate clinical and molecular context.
TERT promoter mutation	May contribute to the molecular diagnosis of glioblastoma, IDH-wildtype.

MGMT promoter methylation has therapeutic implications. MGMT is involved in DNA repair. When its promoter is methylated, MGMT expression is often reduced, making tumour cells potentially more sensitive to alkylating chemotherapy such as temozolomide.

The presence of an IDH mutation also has diagnostic and prognostic implications. An IDH-mutant grade 4 astrocytoma is classified separately from glioblastoma, IDH-wildtype. These distinctions are of crucial importance when interpreting clinical and imaging data.

Therefore, for diagnosis, Brain tumour tissue is obtained through stereotactic needle biopsy or surgical resection is the appropriate diagnostic sample. Histopathological examination, immunohistochemistry and molecular testing are then used to establish the integrated diagnosis. MRI supports the diagnostic process but cannot independently replace tissue diagnosis in the usual clinical setting.

IMAGING FEATURES FOR PREDICTING TREATMENT OUTCOMES:

The Role of MRI in brain tumour surgical planning;

MRI is the preferred imaging technique for assessing the anatomical extent and characteristics of high-grade gliomas. It offers superior soft-tissue contrast compared with CT and allows clinicians to examine tumour characteristics using different imaging sequences.

MRI helps the surgeon to understand where the tumour ends, whether it has infiltrated surrounding tissue, whether it is close to critical neurological structures and how much of it may be removed safely.

A useful starting point is multiparametric MRI, which combines conventional structural sequences with advanced imaging techniques.

Conventional MRI features:

1. **T1-weighted contrast-enhanced imaging**

T1-weighted imaging after gadolinium administration is useful for identifying areas of contrast enhancement.

High-grade gliomas commonly demonstrate irregular or ring-like enhancement surrounding a central region of necrosis.

Contrast enhancement occurs because the blood-brain barrier has become disrupted, allowing gadolinium-based contrast material to enter the extracellular space.

In surgical planning, the enhancing region helps estimate the visible tumour burden. It can also be used to identify regions for targeted biopsy.

However, the enhancing boundary does not necessarily represent the complete extent of viable tumour. Glioma cells may infiltrate surrounding tissue beyond the enhancing region.

2. **T2-weighted imaging**

T2-weighted imaging demonstrates regions containing increased water content.

High-grade gliomas often appear hyperintense on T2-weighted images, particularly where there is surrounding oedema, tumour infiltration or necrosis.

The sequence provides useful information about the wider abnormality surrounding the tumour. However, T2 signal abnormalities are not specific to tumour infiltration, since several other pathological processes can produce similar appearances.

3. Fluid-Attenuated Inversion Recovery (FLAIR)

FLAIR suppresses the signal from cerebrospinal fluid, making abnormalities in the surrounding brain tissue easier to identify.

It is particularly useful for visualising peritumoral hyperintensity.

For instance, a glioblastoma located in the frontal lobe may exhibit an enhancing central mass surrounded by a much larger FLAIR-hyperintense region.

The FLAIR abnormality may contain a combination of vasogenic oedema and infiltrating tumour cells. This distinction matters because oedema may decrease following treatment, while infiltrating tumour cells may remain.

4. Diffusion-weighted imaging and apparent diffusion coefficient

Diffusion-Weighted Imaging (DWI) measures the movement of water molecules within tissue.

Its quantitative counterpart, the Apparent Diffusion Coefficient (ADC), provides information about water diffusion.

Regions with densely packed tumour cells may demonstrate restricted diffusion and relatively low ADC values.

This relationship is useful because tumour cellularity can influence treatment response and disease behaviour.

Nevertheless, low ADC is not specific to high-grade glioma. Other conditions, including ischaemia, infection and some treatment-related changes, can also produce restricted diffusion.

Advanced MRI features;

These offer additional insight into tumour biology.

Imaging technique	Quantitative or qualitative feature	Potential clinical application
Dynamic Susceptibility Contrast (DSC) perfusion	Relative Cerebral Blood Volume (rCBV)	Estimates tumour vascularity and angiogenesis.
Dynamic Contrast-Enhanced (DCE) MRI	Ktrans and other permeability parameters	Characterises vascular permeability and blood-brain barrier disruption.

MR spectroscopy	Choline, NAA, creatine and lactate	Provides information about cellular metabolism.
Diffusion Tensor Imaging (DTI)	Fractional anisotropy and fibre orientation	Assesses white matter tract displacement or disruption.
Functional MRI (fMRI)	Blood oxygen level-dependent activation	Helps identify functional brain regions.

Relative cerebral blood volume;

- High-grade gliomas often exhibit increased vascularity due to tumour angiogenesis.
- DSC perfusion MRI measures changes in signal intensity during the passage of a contrast agent through cerebral blood vessels. The resulting data can be used to estimate relative cerebral blood volume.
- High rCBV values may indicate increased vascularity and can help identify biologically active tumour regions.
- These regions may also be useful targets for stereotactic biopsy because sampling a highly vascular, viable tumour region can sometimes provide more representative tissue than sampling a necrotic centre.
- However, rCBV is influenced by acquisition parameters, contrast leakage and post-processing methods. Measurements should therefore be interpreted cautiously.

Magnetic resonance spectroscopy

MR spectroscopy provides information about the chemical composition of tissue.

Several metabolites are particularly relevant:

- **Choline:** Often increased due to cell membrane synthesis and turnover.
- **N-acetylaspartate (NAA):** Often reduced where normal neuronal tissue is damaged or displaced.
- **Creatine:** Commonly used as a reference metabolite.
- **Lactate:** May increase in regions of altered anaerobic metabolism.

An increased choline-to-NAA ratio may indicate abnormal tumour metabolism.

Spectroscopy can therefore complement conventional MRI, although its spatial resolution and sensitivity to acquisition conditions can limit its clinical application.

Diffusion tensor imaging and functional MRI

One of the most difficult aspects of glioma surgery is determining whether the tumour involves or lies close to critical neurological pathways.

DTI estimates the directionality of water diffusion and can be used to reconstruct white matter fibre pathways.

Functional MRI can help identify areas involved in language, motor and other neurological functions.

For example, a tumour located near the motor cortex may appear relatively small on conventional MRI but still present substantial surgical risk if it lies close to the corticospinal tract.

These imaging techniques can therefore contribute to determining the safest surgical corridor and the maximum extent of resection that can be achieved without unacceptable neurological damage.

They do not replace intraoperative mapping or direct neurosurgical assessment.

MRI features associated with treatment outcomes;

Feature	**Potential relationship with outcomes**	**Relevance to planning**
Total tumour volume	Larger tumour burden may be associated with poorer prognosis and increased treatment complexity.	Estimates the overall extent of disease.
Enhancing tumour volume	Reflects the volume of contrast-enhancing abnormality.	Helps quantify visible tumour burden.
Necrotic fraction	Reflects the proportion of the tumour core containing necrotic tissue.	Helps characterise tumour heterogeneity.
FLAIR abnormality volume	May reflect oedema and infiltrative tumour beyond the enhancing region.	Assesses the wider extent of abnormal tissue.
ADC values	Low values can indicate increased cellularity.	Helps identify regions with different tissue characteristics.
rCBV	Increased values can reflect angiogenesis and vascularity.	Supports identification of biologically active tumour regions.
Tumour location	Involvement of eloquent regions can restrict surgical resection.	Helps assess surgical accessibility and functional risk.
Residual tumour volume	Greater postoperative residual enhancing tumour is generally associated with poorer outcomes.	Assesses the effectiveness of resection.

DTI tract displacement	Indicates the relationship between the tumour and white matter pathways.	Supports surgical corridor selection.

There is a distinction worth maintaining here. Some features are prognostic, while others primarily help the surgeon assess feasibility and risk. Tumour location, for instance, is highly relevant to surgical planning but should not be interpreted as an independent survival predictor without considering other patient characteristics.

Clinical and molecular factors that should accompany MRI features

A model developed to predict treatment outcomes should not rely exclusively on imaging data.

The following variables will also be considered:

1. Patient age.
2. Karnofsky Performance Status (KPS) or another appropriate functional assessment.
3. IDH mutation status.
4. MGMT promoter methylation status.
5. Extent of surgical resection.
6. Treatment received, including radiotherapy and temozolomide.
7. Tumour recurrence status.

For example, two patients may have similarly sized glioblastomas, but their outcomes may differ considerably because of differences in age, functional status, molecular characteristics and treatment response.

A combined radiological, molecular and clinical model is therefore more defensible than a model based only on MRI signal intensity.

Proposed imaging features for the study;

Feature category	Features to extract	Expected purpose
Tumour morphology	Volume, centroid, surface area and shape	Quantify tumour geometry.
Tumour composition	Enhancing, necrotic and oedematous fractions	Describe tumour heterogeneity.
MRI intensity	Mean, median, standard deviation and percentiles	Characterise regional signal distribution.
Spatial characteristics	Tumour centroid and anatomical location	Support anatomical interpretation.
Diffusion	Mean and median ADC, where available	Estimate diffusion characteristics.

Perfusion	Mean rCBV and upper-percentile rCBV, where available	Characterise tumour vascularity.
Functional anatomy	Distance from relevant white matter tracts and functional regions	Support assessment of surgical risk.

The initial implementation will concentrate on features that can be extracted directly from BraTS MRI data. ADC and rCBV measurements will be treated as optional extensions because the standard BraTS four-modality input does not consistently include dedicated ADC and perfusion maps.

MRI Image Segmentation and Feature Extraction

- Image Data Source: BraTs: Medical Segmentation Decathlon (MSD), Task01: Brain Tumours
- Programming Languages Employed: Python, R
- Packages: Python (nibabel, numpy, pandas, matplotlib, scipy, scikit-learn, scikit-image) R(readr, ggplot2, dplyr, tidyr)

While the dataset obtained for this assignment is a section of the Brain Tumour Segmentation (BraTs) data set (Task 01: Brain Tumours), an interpretation of the significance of the segmentation and feature extraction is performed using the sample image data BraTs_001.nii.gz, which is an MRI capture of a brain glioma.

The features examined are therefore those obtainable via conventional MRI but not advanced techniques of functional MRI. The dataset provides a vast collection of 45 variables from which distinct clinical insights are obtainable. However, for this assignment, I will focus on the following features, to provide clinical insight in my capacity;

Measure/Feature	**Value**
Whole Tumour (WT) volume	111.724 cm^3
Tumour Core (TC) volume	58.674 cm^3
Enhancing Tumour volume	31.485 cm^3
Non-Enhancing/Necrotic Core volume	= (TC - ET) = 27.189 cm^3
Oedema/Infiltrated Tissue Volume	= $[{}^{47}/_{100} \times WT]$ = 52.510 cm^3
Components	2
Bounding Box	55×96×74

Intuition from Extracted Features and Segmentation:

On the T1ce (T1-Weighted Contrast Enhanced) slice, a thick, irregular enhancing rim surrounds a dark necrotic centre, and a band of oedema extends away from the mass. This is indicative of the suspected high-grade glioma.

However, since Abscess, metastasis and lymphoma can look similar when imaged, tissue and molecular testing are essential to support conclusive diagnosis.

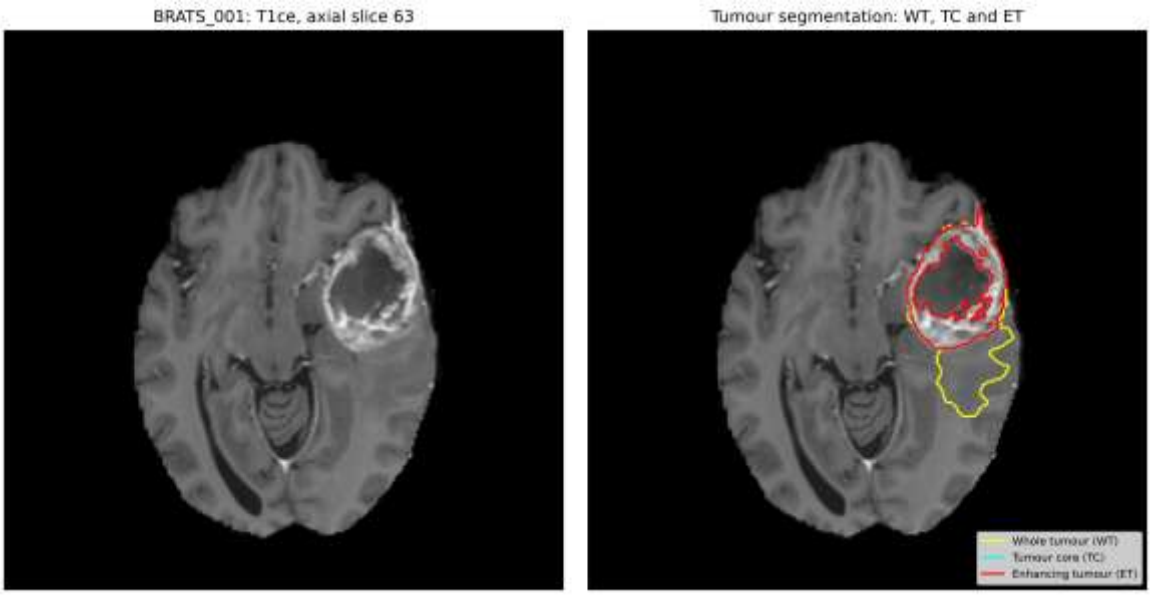


Figure 1: Plot of Whole Tumour (WT), Tumour Core (TC) and Enhancing Tumour (ET) for image sample BRATS_001. (The Cyan TC is hidden beneath the red ET.)

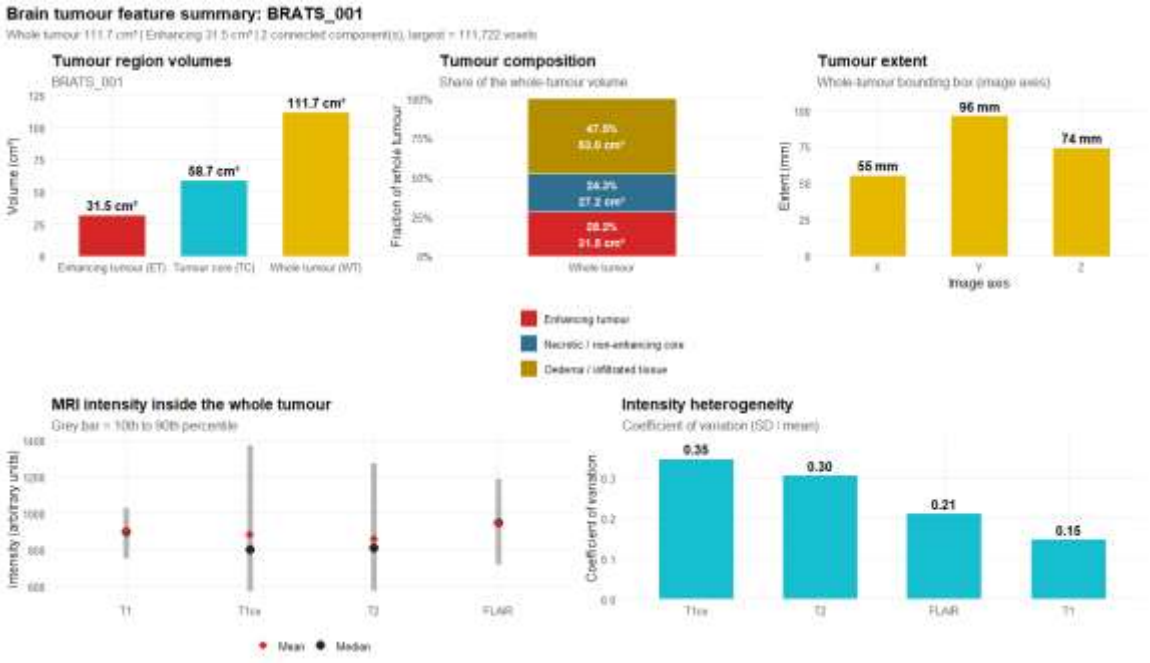


Figure 2: Dashboard of Brain Tumour summary for BRATS_001

Contribution to Surgical Planning Procedure;

1. **The enhancing volume is the baseline for extent of resection.**

The 31.5 cm³ figure is what postoperative MRI will be compared against. Residual tumour volume matters prognostically: the RANO resect group found that lower absolute residual

tumour volumes were linked to better outcomes, and maximal contrast-enhancing resection beat submaximal resection or biopsy. The same group found that when the enhancing tumour is completely removed, also removing most of the non-enhancing tumour (leaving ⩽5 cm^3) gives further survival benefit, which defines "supramaximal" resection. Here the oedema region is a candidate for that, since it may contain infiltrating tumour. It should only be pursued where it is safe.

2. **The large core means sampling and targeting matter.**

About half of the core is necrotic, and necrotic tissue is often non-diagnostic. Biopsy or resection samples should come from the viable enhancing rim.

3. **The size raises perioperative risks.**

A 111 cm^3 lesion with extensive oedema suggests possible raised intracranial pressure and mass effect, so steroid use, close neurological monitoring and a check for midline shift or herniation signs should be planned.

4. **Location drives the risk.**

On this slice the lesion sits anterolateral and lateral, in the temporal or fronto-temporal region near the sylvian fissure and insula, with oedema tracking posteriorly. If that is confirmed, a neurosurgeon would check for:

- language areas and white-matter tracts (arcuate and inferior fronto-occipital fasciculi) if it is in the dominant hemisphere;
- the optic radiation (Meyer's loop) in the temporal lobe;
- the corticospinal tract and lenticulostriate vessels near the insula and internal capsule;
- middle cerebral artery branches in the sylvian fissure.

Conclusion:

Comprehensive Surgical planning for high-grade glioma is a complex process which must involve a series of examinations of which MRI imaging and AI/ML techniques for segmentation and feature extraction are only but an aiding part in the broader process.

Advanced Imaging procedures, as well as histopathological study results from obtained biopsies must be considered to come up with a definitive conclusion on the suitability and feasibility of the surgical procedure and level of risk it bears upon the patient.